If f is bounded on a rectangle R, then ----------

149page

Answer (Please select your correct option)

- $\overline{\int_R} f(X)d(X) \leq \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) \geq \underline{\int_R} f(X)d(X)$
- None of these

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If f is unbounded on the nondegenerate rectangle R in $\mathbb{R}^2$, then———

page145

Answer (Please select your correct option)

- f is not integrable on R
- f is integrable on R
- f is not differentiable on R
- None of these

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A bounded function f is integrable on a rectangle R if and only if ----------

$$\overline{\int_R} f(X)d(X) = \underline{\int_R} f(X)d(X)$$

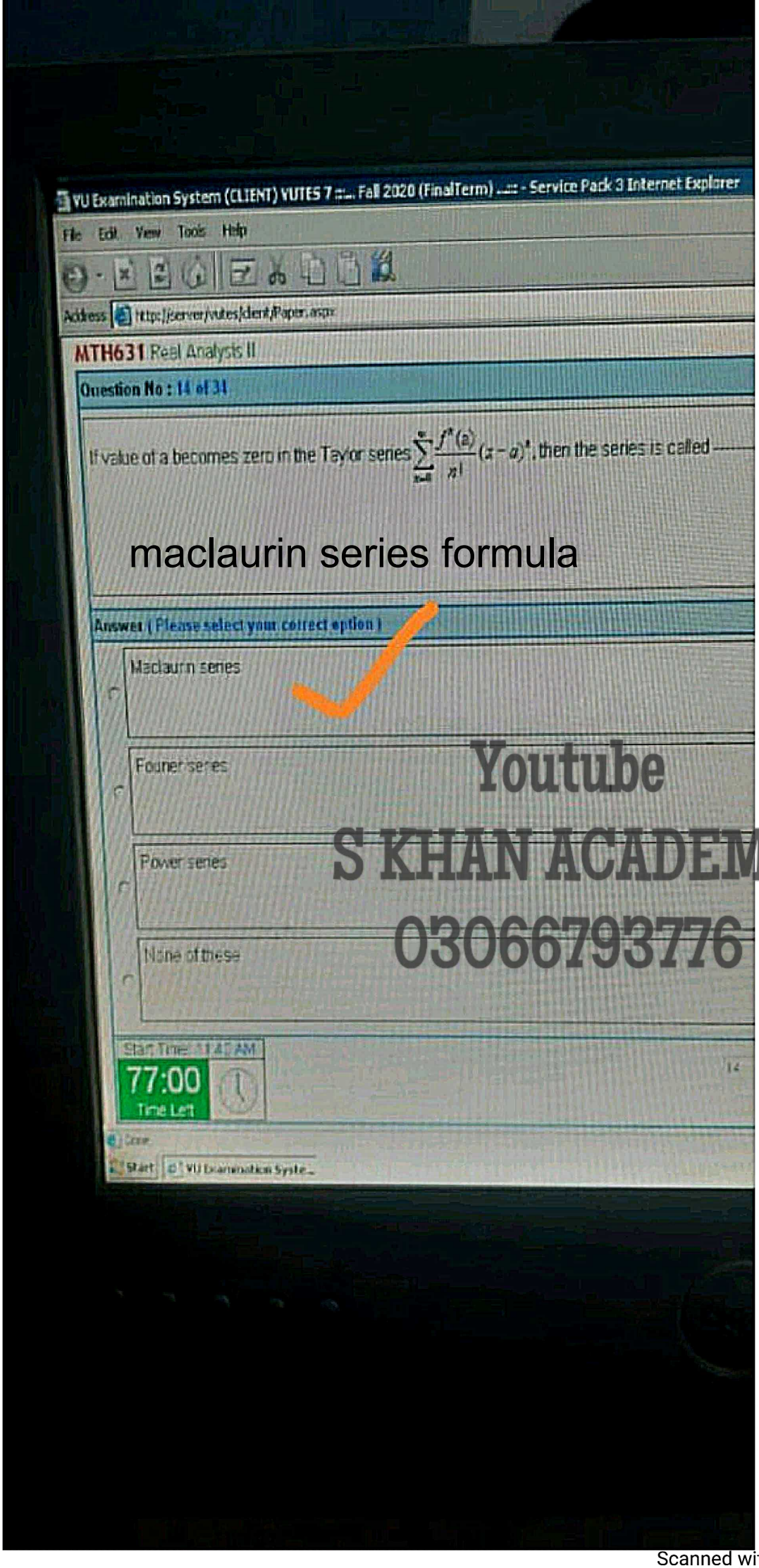
Answer (Please select your correct option)

- $\overline{\int_R} f(X)d(X) = \underline{\int_R} f(X)d(X)$ page150
- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) < \underline{\int_R} f(X)d(X)$
- None of these

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An absolutely convergent integral is always ----------

Answer (Please select your correct option)

- divergent
- convergent
- conditionally convergent
- None of these

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If f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|dx < \infty$, then $\int_a^b f(x)dx$ is ----

Answer (Please select your correct option)

- divergent
- conditionally convergent
- convergent ✓
- None of these

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Question No : 27 of 34

The ---------- of finitely manay sets with zero content has zero content

Answer (Please select your correct option)

- complement
- union
- difference
- None of these

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A set of points of the form $X = X_0 + tU, t_1 \leq t \leq t_2$ is called .....

Answer (Please select your correct option)

- the neighbourhood.
- line segment. ✓
- the closure.
- None of these

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38:00 Time Left

30

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22

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_____ is the only region in R.

Answer (Please select your correct option)

- Set of finite points of R
- Interval
- Any finite sequence of R
- None of these

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Question Summary : (Attempted Question)

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If H is an open covering of a compact subset S then, S can be covered by ———

Answer (Please select your correct option)

- infinitely many sets from H
- finitely many sets from H ✓
- atleast one set from H
- None of these

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2

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 4 of 34

For the function $f(x,y)=\frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y)\to(0,0)$ along the line $y=x$ is ----

Answer (Please select your correct option)

- $\frac{1}{2}$
- $-\frac{1}{2}$
- 0
- 1

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Question Summary : (Attempted Question)

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Question No : 1 of 34

The graph of an odd function is symmetrical about ----------

Answer (Please select your correct option)

- x-axis
- y-axis
- origin ✓
- None of these

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Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17

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MTH631 Real Analysis II

Question No : 5 of 34

If f and g are continuous functions at the point X_0 in R^n, then $\frac{f}{g}$ is continuous at X_0 provided that ——

Answer (Please select your correct option)

- $g(X_0) = 0$
- $g(X_0) \neq 0$
- $f(X_0) = g(X_0) = 0$
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 21 of 34

Suppose f is a continuous and $F(x) = \int_a^x f(t)dt$ is bounded on $[a,b)$. Let g' be an

Answer (Please select your correct option)

- $\int_a^b f(x)g(x)dx$ converges
- $\int_a^b f(x)g(x)dx$ diverges
- $\int_a^b f(x)g(x)dx$ conditionaly converges
- None of these

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MTH631 Real Analysis II

Question No : 21 of 34

Suppose f is a continuous and $F(x) = \int_a^x f(t)dt$ is bounded on $[a,b)$. Let g' be an

Answer (Please select your correct option)

- $\int_a^b f(x)g(x)dx$ converges
- $\int_a^b f(x)g(x)dx$ diverges
- $\int_a^b f(x)g(x)dx$ conditionally converges
- None of these

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Question No : 24 of 34

If f is bounded on a rectangle R, then ----------

Answer (Please select your correct option)

- $\overline{\int_R} f(x)d(x) \leq \underline{\int_R} f(x)d(x)$
- $\overline{\int_R} f(x)d(x) > \underline{\int_R} f(x)d(x)$
- $\overline{\int_R} f(x)d(x) \geq \underline{\int_R} f(x)d(x)$
- None of these

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(19) An absolutly convergent integral is alway ⟶
(i) divergent (ii) convergent (iii) conditionally convergent
(iv) none of thes-

(22) If f is unbounded on the non degenerate rectangle R in R^n then —
(i) f is not integrable on R (ii) f is integrable on R
(iii) f is not differentiable on R (iv) None of th

(24) The lower integral $\underline{\int_R} f(x)dx$ of f over R is the Supremum of all the ——
(i) lower Sums (ii) upper Sums (iii) average Sums
None of thes

(27) The — of finitely manay sets with zero content has zero content
(i) Complement (ii) Union (iii) differece (iv) Non of thes

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Course Code: _____

(30) A set is closed if and only if it contains all of its
(i) Neighborhoods (ii) Limit Pts (iii) Exteriors
(iv) None of these

(28) A ___ surface in R^n has zero content
(i) Parabolic (ii) elliptical (iii) differentiable
(iv) None of these.

The graph of an even function is symmetrical about origin.

2) If H is an open covering of a compact subset S.
Then S can be covered by atleast on set from H

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Q) The greatest lower bound of a set is called ____ of the set.
(i) Supremum (ii) infimum (iii) greatest lower bound
(iv) None of these

Q) If $f_{x_1}, f_{x_2}, \ldots f_{x_n}$ exist on a neighborhood of x_0 and are continuous at x_0, the f is ____ at x_0.
(i) uniformly convergent (ii) Piecewise convergent
(iii) differentiable (iv) None of these

Q) A function of several variables may have first order partial derivatives at a Pt x_0 but ____ continuous at x_0.
(i) fail to be (ii) must be (iii) always not (iv) None of these

Q) if $f(x) = 3x + 5x^3$ then at $x=1$ $df =$ ____
18dy, 18dx, 18, None of These

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Campus Code: _____ Student Signature: _____

Course Code: _____ Student ID: _____

(13) if $h = f(r, \theta)$ then using the chain rule $\frac{\partial h}{\partial r} =$ _____

(i) $\frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial y}$ (ii) $\frac{\partial f}{\partial x} + \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} + \frac{\partial y}{\partial r}$

(iii) $\frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial r}$ (iv) non of thes.

(14) $h = f(x, y)$ $x = r\cos\theta$ and $y = r\sin\theta$ where then using chain rule $\frac{\partial h}{\partial r} =$ _____

(i) $\cos\theta \frac{\partial f}{\partial \theta} + \sin\theta \frac{\partial f}{\partial r}$ (ii) $\cos\theta \frac{\partial f}{\partial x} + \sin\theta \frac{\partial f}{\partial y}$

(iii) $\cos\theta \frac{\partial f}{\partial x} - \sin\theta \frac{\partial f}{\partial y}$ None of the

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Question No : 26 of 41

If H has 6 elements, then all cosets of H must have _____ elements.

Answer (Please select your correct option)

- 3
- 9
- 12
- 6

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 2

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Question No : 3 of 34

is the only region in R

Answer (Please select your correct option)

- Set of finite points of R
- Interval ✓
- Any finite sequence of R
- None of these

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Question Summary : (Attempted Question)

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If $h = f(x,y), x = f(r,\theta)$ and $y = f(r,\theta)$, then using the chain rule, $\frac{\partial h}{\partial \theta}$ = ...

Answer (Please select your correct option)

- $\frac{\partial f}{\partial r}\cdot\frac{\partial r}{\partial \theta}+\frac{\partial f}{\partial r}\cdot\frac{\partial r}{\partial \theta}$
- $\frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial \theta}+\frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial \theta}$ ✓
- $\frac{\partial f}{\partial x}+\frac{\partial x}{\partial \theta}+\frac{\partial f}{\partial y}+\frac{\partial y}{\partial \theta}$
- None of these

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13

Question Summary : (Attempted Question)

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MTH031 Real Analysis II

Question No : 14 of 34

If $h = f(x,y), x = r\cos\theta$ and $y = r\sin\theta$, where then using the chain rule, $\frac{\partial h}{\partial r}$ = ...

Answer (Please select your correct option)

- $\cos\theta\frac{\partial f}{\partial \theta} + \sin\theta\frac{\partial f}{\partial r}$
- $\cos\theta\frac{\partial f}{\partial x} + \sin\theta\frac{\partial f}{\partial y}$ ✓
- $\cos\theta\frac{\partial f}{\partial x} - \sin\theta\frac{\partial f}{\partial y}$
- None of these

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14

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

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MTH631 Real Analysis II

Question No : 9 of 34

A point on the graph of a function at which the tangent line is parallel to x-axis is called the .....

Answer (Please select your correct option)

- critical point ✓
- point of inflection
- point of relative extrema
- None of these

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TH631 Real Analysis II

Question No : 5 of 34

If f and g are continuous functions at the point X_0 in R^n, then $\frac{f}{g}$ is continuous at X_0 provided that ——

Answer (Please select your correct option)

- $g(X_0) = 0$
- $g(X_0) \neq 0$ ✓
- $f(X_0) = g(X_0) = 0$
- None of these

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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_____ is the only region in R.

Answer (Please select your correct option)

- Set of finite points of R
- Interval ✓
- Any finite sequence of R
- None of these

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3

Question Summary : (Attempted Question)

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ATH631 Real Analysis II

Question No : 4 of 34

For the function $f(x,y)=\frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y)\to(0,0)$ along the line $y=x$ is ---

Answer (Please select your correct option)

- $\frac{1}{2}$ ✓
- $-\frac{1}{2}$
- 0
- 1

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MTH631 Real Analysis II

Question No : 2 of 34

If H is an open covering of a compact subset S then, S can be covered by ------

Answer (Please select your correct option)

- infinitely many sets from H
- finitely many sets from H ✓
- atleast one set from H

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If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \leq 0$ for all $X \neq X_0$. Then p is called ———

Answer (Please select your correct option)

- positive semidefinite
- positive definite
- negative semidefinite ✓
- negative definite

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A function of the form $L(X) = f_{x_1}(X_0)x_1 + f_{x_2}(X_0)x_2 + \ldots + f_{x_n}(X_0)x_n$ is called ———

Answer (Please select your correct option)

- quadractic function
- differential of f at X_0
- constant function
- None of these

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12

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22

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If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \leq 0$ for all $X \neq X_0$. Then p is called ———

Answer (Please select your correct option)

- positive semidefinite
- positive definite
- negative semidefinite ✓
- negative definite

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15

Question Summary : (Attempted Question)

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If $h = f(x,y), x = f(r,\theta)$ and $y = f(r,\theta)$, then using the chain rule, $\frac{\partial h}{\partial \theta}$ = ...

Answer (Please select your correct option)

- $\frac{\partial f}{\partial r}\cdot\frac{\partial r}{\partial \theta}+\frac{\partial f}{\partial r}\cdot\frac{\partial r}{\partial \theta}$
- $\frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial \theta}+\frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial \theta}$ ✓
- $\frac{\partial f}{\partial x}+\frac{\partial x}{\partial \theta}+\frac{\partial f}{\partial y}+\frac{\partial y}{\partial \theta}$
- None of these

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Question Summary : (Attempted Question)
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MTH631 Real Analysis II

Question No : 11 of 34

If $h = f(x,y), x = r\cos\theta$ and $y = r\sin\theta$, where then using the chain rule, $\frac{\partial h}{\partial r}$ = ...

Answer (Please select your correct option)

- $\cos\theta\frac{\partial f}{\partial \theta} + \sin\theta\frac{\partial f}{\partial r}$
- $\cos\theta\frac{\partial f}{\partial x} + \sin\theta\frac{\partial f}{\partial y}$ ✓
- $\cos\theta\frac{\partial f}{\partial x} - \sin\theta\frac{\partial f}{\partial y}$
- None of these

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Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 18 of 34 Marks:

A function f is said to be absolutely integrable on $[a,b)$ if f is locally integrable on $[a,b)$ and ----------

Answer (Please select your correct option)

- $\int_a^b |f(x)|dx < \infty$ ✓
- $\int_a^b |f(x)|dx = \infty$

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ATH631 Real Analysis II

Question No : 18 of 34 Marks:

A function f is said to be absolutely integrable on $[a,b)$ if f is locally integrable on $[a,b)$ and ----------

Answer (Please select your correct option)

- $\int_a^b |f(x)|dx < \infty$
- $\int_a^b |f(x)|dx = \infty$
- $\int_a^b |f(x)|dx \geq \infty$
- None of these

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18

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26

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MTH631 Real Analysis II

Question No : 19 of 34

An absolutely convergent integral is always ----------

Answer (Please select your correct option)

- divergent
- convergent ✓
- conditionally convergent
- None of these

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19

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 20 of 34

If f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|dx < \infty$, then $\int_a^b f(x)dx$ is ----------

Answer (Please select your correct option)

- divergent
- conditionally convergent
- convergent ✓
- None of these

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20

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

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MTH631 Real Analysis II

Marks: 10 (Budgeted Time Min)

Question No : 34 of 34

Suppose $U_0 \in T$ is a limit point of T, g is continuous at U_0 and f is continuous at $X_0 = G(U_0)$. Then, show that $f \circ g$ is continuous at U_0.

Answer (Please click here to Add Answer)

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

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MTH631 Real Analysis II

Question No : 33 of 34 **Marks: 5 (Budgeted Time M**

Show that $(0,0)$ is a critical point of the function $p(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$ and it has positive semidefinite second differentials at $(0,0$

Answer (Please click here to Add Answer)

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33

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 32 of 34 Mark

Define a locally integrable function. Also show that the function $f(x) = \log x$ is locally integrable on $[0, 1)$.

Answer (Please click here to Add Answer)

100%

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

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